

Translated

From: e-maxx.ru

Sprague-Grundy theorem. Nim

Introduction

This theorem describes the so-called **impartial** two-player game, i.e. those in which the available moves and winning/losing depends only on the state of the game. In other words, the only difference between the two players is that one of them moves first.

Additionally, we assume that the game has **perfect information**, i.e. no information is hidden from the players (they know the rules and the possible moves).

It is assumed that the game is **finite**, i.e. after a certain number of moves, one of the players will end up in a losing position — from which they can't move to another position. On the other side, the player who set up this position for the opponent wins. Understandably, there are no draws in this game.

Such games can be completely described by a *directed acyclic graph*: the vertices are game states and the edges are transitions (moves). A vertex without outgoing edges is a losing vertex (a player who must make a move from this vertex loses).

Since there are no draws, we can classify all game states as either **winning** or **losing**. Winning states are those from which there is a move that causes inevitable defeat of the other player, even with their best response. Losing states are those from which all moves lead to winning states for the other player. Summarizing, a state is winning if there is at least one transition to a losing state and is losing if there isn't at least one transition to a losing state.

Our task is to classify the states of a given game.

The theory of such games was independently developed by Roland Sprague in 1935 and Patrick Michael Grundy in 1939.

Nim

This game obeys the restrictions described above. Moreover, *any* perfect-information impartial two-player game can be reduced to the game of Nim. Studying this game will allow us to solve all other similar games, but more on that later.